

BMJ Open Exposure to school-based tobacco prevention interventions in low-income and middle-income countries and its association with psychosocial predictors of smoking among adolescents: a pooled cross-sectional analysis of Global Youth Tobacco Survey data from 38 countries

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ABSTRACT

Objectives To describe the prevalence of school-based tobacco prevention programme exposure among adolescents in low-income and middle-income countries (LMICs) and its association with psychosocial predictors of smoking.

Design Analysis of pooled cross-sectional data.

Setting Global Youth Tobacco Survey (GYTS), conducted in 38 LMICs.

Participants This was a pooled analysis of data involving a total of 132 755 adolescent respondents to GYTS in 38 LMICs across Africa, Europe and Central/South America between 2014 and 2017.

Exposure and outcome measures The primary independent variable for this study was self-reported exposure to school-based tobacco prevention programmes in the past year. Five psychosocial determinants of smoking were explored as outcomes: perceived addictiveness of nicotine, perceived harm of secondhand smoke exposure, support for restricting cigarette smoking at public indoor locations, support for restricting cigarette smoking at public outdoor areas and self-reported prediction of enjoying cigarette smoking. Multivariable logistic regression models were used to examine the relationship between exposure to school-based tobacco prevention programmes and study outcomes, controlling for sociodemographic and smoking-related characteristics of respondents.

Results Overall, 59.1% of adolescents in LMICs self-reported exposure to school-based tobacco prevention programmes. The country-specific prevalence of adolescent exposure to school-based tobacco interventions ranged from 24.9% in the Comoros to 99.3% in Turkmenistan. Exposure to school-based tobacco interventions was significantly associated with greater secondhand smoke harm perceptions (adjusted OR (AOR): 1.69; 95% CI: 1.69 to 1.70), perceptions of addictiveness (AOR: 1.37; 95% CI: 1.36 to 1.37) and supporting tobacco use restrictions indoors (AOR: 1.70; 95% CI: 1.69 to 1.70)

STRENGTHS AND LIMITATIONS OF THIS STUDY

- ⇒ This study includes a large sample of adolescents from 38 low-income and middle-income countries (LMICs) across Africa, Europe and Central/South America, which significantly strengthens the statistical power and robustness of our analyses.
- ⇒ The study used weighting to account for the complex survey weights used in the Global Youth Tobacco Survey to obtain estimates that are representative of the national populations.
- ⇒ Given that the sampled countries are not representatives of all LMICs in corresponding WHO regions, findings may not be generalisable to the respective regions.
- ⇒ The study did not account for the variation in the structure of school-based tobacco prevention educational programmes, as well as grade forms across the different LMICs.
- ⇒ The study is prone to recall bias as respondents had to recollect past behaviours and exposures to respond to survey questions.

and outdoors (AOR: 1.59; 95% CI: 1.59 to 1.60). Exposure to school-based tobacco interventions was associated with lower odds of anticipating enjoying cigarette smoking (AOR: 0.76; 95% CI: 0.76 to 0.76).

Conclusion Exposure to tobacco prevention programmes in schools is suboptimal in LMICs. Given the protective associations described in this study from school-based tobacco prevention programme exposure, it is imperative that national governments implement school-based programmes into ongoing tobacco control measures.

INTRODUCTION

Cigarette smoking remains one of the leading causes of preventable deaths globally.¹ While

achieving optimal tobacco control remains a significant challenge for most countries around the globe, low-income and middle-income countries (LMICs), defined by the World Bank as countries with a gross national income per capita between US\$1046 and US\$4095, face even greater challenges. The tobacco industry continues to actively recruit new cigarette users in LMICs by applying aggressive marketing tactics including those that have been banned in developed countries, such as free product giveaways and the sale of cigarettes in single sticks.^{2–4} Consequently, high protobacco advertisement exposure rates have been described among adolescents in Sub-Saharan Africa, as well as remarkably young ages at smoking initiation.⁵ Given the health and economic ramifications of cigarette smoking, effective counter-measures must be implemented in LMICs to reduce cigarette smoking among adolescents.

Psychosocial determinants of cigarette smoking, including harm perception from nicotine addictiveness and secondary exposure to cigarette smoke, are modifiable factors that could be targeted by interventions to reduce cigarette use among adolescents. More so, social acceptability, such as public support for tobacco control policies (eg, restrictions on smoking in public indoor and outdoor spaces), correlates with perceived harm from smoking and influences cigarette use.^{6,7} Along with a high smoking prevalence, several studies have shown school-going adolescents in LMIC to be susceptible to smoking initiation.^{8–10} As such, school-based smoking prevention programmes have been developed to curb tobacco use in this population. Several studies have shown that school-based smoking prevention programmes effectively prevent adolescent cigarette smoking and decrease susceptibility to smoking initiation.^{8–11,13} A recent meta-analysis found that school-based prevention programmes effectively reduced adolescent cigarette smoking initiation by 12% over 1 year.¹¹ Furthermore, adopting multifaceted tobacco control strategies per the WHO's Framework Convention on Tobacco Control (FCTC) within schools' jurisdictions can complement school-based prevention programmes and boost their effectiveness.^{11–14} The FCTC is a tobacco control treaty ratified by 182 member states with a goal of implementing tobacco control measures outlined by the convention framework.

Effective school-based tobacco prevention programmes are theory driven and focused on addressing the psychosocial determinants of tobacco use.^{11–13,15–18} For example, social cognitive theory (SCT) posits that behaviour is influenced by both environmental factors (eg, social influence) and individual determinants (eg, outcome expectations; harm perceptions).^{18,19} SCT is the theoretical foundation for many school-based cigarette smoking prevention programmes that effectively address determinants and antecedents to adolescent cigarette initiation.^{11,12} As such, research that assesses the relationship between exposure to school-based prevention programmes and the psychosocial determinants of adolescent cigarette smoking may inform theory-driven interventions.

In a study of the Global Youth Tobacco Survey (GYTS) in 1999, the proportion of students who reported having been taught in school about the dangers of tobacco use ranged from over 70% in China to approximately 33% in Barbados, Costa Rica, Russian (Moscow), South Africa and Zimbabwe.²⁰ Another study of the 2005–2011 GYTS of 43 countries found that the proportion of students exposed to classes on the danger of smoking ranged from 31.4% (Georgia) to 83.4% (Papua New Guinea).²¹ Despite the varying prevalence of school-based tobacco prevention programmes, research has found that the implementation of these programmes in LMICs are highly fragmented; more so, extensive evaluations of the effectiveness of these programmes have not been conducted.¹⁵ To our knowledge, only a few studies have examined the prevalence of school-based smoking prevention programmes and their association with psychosocial determinants of adolescent tobacco use across a multinational sample of adolescent residents of LMICs. As such, research is needed to examine the reach and impact of school-based prevention programmes in LMICs.

Therefore, in this study, we examined the prevalence of school-based cigarette smoking prevention programmes in 38 LMICs across Africa, Europe and the Americas, from 2014 to 2017. Furthermore, we evaluated the association between exposure to school-based cigarette smoking prevention programmes and five psychosocial determinants of cigarette smoking among adolescents in 38 LMICs. The five psychosocial determinants of cigarette smoking examined were: perceived addictiveness of nicotine, perceived harm of secondhand smoke exposure, support for restricting cigarette smoking at public indoor locations, support for restricting cigarette smoking at public outdoor areas and self-reported prediction of enjoying cigarette smoking (ie, positive outcome expectations). We hypothesise that exposure to school-based cigarette smoking prevention programmes will be associated with greater perceived addictiveness of nicotine, perceived harm of secondhand smoke exposure, and support for restricting cigarette smoking at public indoor and outdoor locations. Additionally, we hypothesise that exposure to school-based cigarette smoking prevention programmes will be associated with lower odds of positive outcome expectations towards cigarette smoking.

METHODS

Study design, data source and population

This study pooled cross-sectional data from the GYTS completed by 38 LMICs across Africa, Europe and the Americas from 2014 to 2017. The GYTS is a school-based survey designed to monitor tobacco use risk factors and behaviours among adolescents in LMICs.²² GYTS is conducted sporadically, and uniformly recent data is unavailable for most countries. Only LMICs that had nationally representative (as opposed to subnational data) and recent data (GYTS conducted on or after 2014) were included in our study. At the time of initiating

data analysis, the most recent data obtainable from the GYTS for most LMIC ranged between 2014 and 2017. As such, 38 countries that met eligibility were included in the study. The countries included in this study are listed in online supplemental table 1. The GYTS methodology uses a two-stage cluster sample to produce representative samples of students in grades associated with adolescents aged 13–15 years. Each sampling frame includes all public and private schools in a geographic area of interest. A full description of the methodology of the GYTS has been described previously.²² All selected students in the class on the day of survey administration are eligible to participate in GYTS. Participation is anonymous and voluntary using a self-administered questionnaire. This study examined the total sample of adolescents who completed the GYTS within each study country. A total of n=132 755 adolescents completed the GYTS surveys from 2014 to 2017 across all countries included in this study. Country-specific sample sizes are listed in online supplemental table 1. The response rates of GYTS in the different countries included in our study ranged from 79% in the Dominican Republic to 97% in Turkmenistan. The study was reported following the Strengthening The Reporting of Observational Studies in Epidemiology guidelines.²³ Passive parental consent was used for the GYTS. Parents of adolescents received a notification form or letter prior to the survey and were allowed to opt-out.

Measures

Independent variable

School-based tobacco prevention programme exposure

The independent variable for this study was self-reported exposure to school-based tobacco prevention programmes in the past year. Participants were asked: during the past 12 months, were you taught in any of your classes about the dangers of tobacco use? Possible responses to this question were yes, no and I don't remember. For this study, participants who reported 'I don't remember' were categorised as not being exposed to school-based tobacco prevention programmes. As such, this independent variable was dichotomous; no/I don't remember (referent) and yes.

Dependent variables

Psychosocial determinants

This study examined five psychosocial determinants of adolescent tobacco use. First, participants were asked the following two questions related to tobacco harm perceptions: (1) do you think the smoke from other people's tobacco smoking is harmful to you? and (2) once someone has started smoking tobacco, do you think it would be difficult for them to quit? Possible responses for these questions were coded into the following categories: definitely/probably not (referent) and probably/definitely yes.

Next, participants were asked the following two questions related to social norms: are you in favour of banning smoking inside enclosed public places (such as schools,

shops, restaurants, shopping malls, movie theatres, cybercafés)? and (2) are you in favour of banning smoking at outdoor public spaces (such as playgrounds, sidewalks, entrances to buildings, stadiums, beaches, bus stops)? Possible responses to these questions were no (referent) and yes.

And finally, participants were asked the following question related to outcome expectations: do you agree or disagree with the following: 'I think I might enjoy smoking a cigarette'. Possible responses to this question were coded into the following categories: strongly agree/agree (referent) and disagree/strongly disagree. An additional response to this question was, I currently smoke cigarettes. Participants who provided this response were categorised into the referent outcome.

Covariates

Demographic characteristics

Age was coded into three categories: 12 years or less (referent), 13–15 years old and greater than 15 years. Sex was coded dichotomously; males served as the referent category.

Self-reported exposure to tobacco marketing was assessed by developing a cumulative exposure variable coded continuously from 0 to 4, based on the responses to the following questions: (1) during the past 30 days, did you see any people using tobacco on TV, in videos or in movies?; (2) during the past 30 days, did you see any advertisements or promotions for tobacco products at points of sale (such as stores, shops, kiosks, supermarkets, etc)?; (3) do you have something (eg, t-shirt, pen, backpack) with a tobacco product brand logo on it? and (4) has a person working for a tobacco company ever offered you a free tobacco product? Possible responses for each question were yes (coded as 1) and no (coded as 0); a sum of these responses was computed for each participant to create the cumulative exposure variable.

Self-reported exposure to secondhand smoking in any location was assessed based on the response to the following survey questions: (1) during the past 7 days, on how many days has anyone smoked inside your home, in your presence?; (2) during the past 7 days, on how many days has anyone smoked in your presence, inside any enclosed public place, other than your home (such as school, shops, restaurants, supermarkets, movie theatres, cybercafés? and (3) during the past 7 days, on how many days has anyone smoked in your presence, at any outdoor public place (such as playgrounds, sidewalks, entrances to buildings, bus stops, beaches)? Participants who responded other than '0 days' to any of the three questions were considered to be exposed to secondhand smoke.

Self-reported exposure to antitobacco marketing was analysed as a continuous variable as a composite of the following three questions: 'During the past 30 days, did you: (1) see or hear any antitobacco media messages on television, radio, internet, billboards, posters, newspapers, magazines or movies?; (2) see or hear any



antitobacco messages at sports events, fairs, concerts, community events or social gatherings? and (3) see any health warnings on cigarette packages? (eg, tobacco seriously damages health, tobacco kills). Responses for these questions varied slightly; however, those who responded 'yes' were coded as '1' and counted towards the composite number of antitobacco messaging channels of exposure.

Self-reported ever cigarette smoking was assessed based on the question: have you ever tried or experimented with cigarette smoking, even one or two puffs?

Data analysis

Five multilevel, multivariate logistic regression models were used to test the study hypotheses. For all statistical tests, self-reported exposure to school-based tobacco prevention programmes was the primary independent variable. A single multilevel, multivariate logistic regression model was used for each listed outcome. For all analyses, sex, age category, cumulative tobacco marketing exposure, cumulative secondhand smoke exposure and current cigarette smoking served as fixed-effect covariates. Additionally, country was used as a random effect (ie, nesting variable) to generate multilevel models,

which accounted for the nesting of participants within study countries. Inclusion of this random-effect variable allowed our models to control for country-level variance in potential confounding variables such as tobacco regulatory policy and enforcement. Data were weighted to be representative of adolescents in the 38 countries included in this study, and 99% CIs were calculated to account for the large sample size. Estimates were calculated using a fitted starting value based on a simplified model.²⁴ All analyses were conducted using STATA V.14.2 (College Station, Texas, USA).

Patient and public involvement

None.

RESULTS

A total of 132 755 adolescents from 38 LMICs in Africa (37.8%), Europe (31.0%) and the Americas (31.2%) were included in this study (table 1).

The overall study sample was evenly distributed by sex (50.2% girls; 49.8% boys), while 10.2% were 12 years or young, 69.2% were 13–15 years old and 20.6% were 16

Table 1 Description of sample characteristics by LMIC region (n=132 755; GYTS 2014–2017)

Characteristics	Full sample	Region		
		Africa	Europe	Americas
		Per cent of sample		
		37.80%	31.00%	31.20%
Sex				
Male	49.80%	49.40%	50.50%	49.70%
Female	50.20%	50.60%	49.50%	50.30%
Age category				
≤12 years	10.20%	11.10%	7.80%	11.50%
13–15 years	69.20%	70.50%	78.80%	58.20%
>15 years	20.60%	18.50%	13.50%	30.30%
Marketing exposure*				
Mean:	0.92	0.84	0.97	0.98
Antitobacco marketing exposure†				
Mean:	1.24	1.08	1.37	1.27
Secondhand smoke exposure‡				
Yes	51.70%	51.90%	61.30%	42.00%
Ever cigarette smoking				
Yes	19.4%	8.9%	30.7%	21.6%
School-based prevention§				
Yes	59.10%	57.10%	64.40%	56.40%

*Composite score (0–4) of self-reported exposure to four channels of tobacco marketing: (1) TV, movies, videos; (2) point of sale; (3) branded items and (4) offered a free tobacco product by the tobacco industry.
†Composite score (0–3) of self-reported exposure to antitobacco marketing via: (1) media; (2) events and (3) cigarette packages.
‡Self-reported exposure to secondhand smoke in the past 7 days via one or more of the following locations: (1) at home; (2) indoor public setting and (3) outdoor public setting.
§Self-reported exposure to school-based tobacco prevention programme in the past 12 months.
GYTS, Global Youth Tobacco Survey; LMIC, low-income and middle-income country.

Table 2 Prevalence of exposure to school-based smoking prevention programmes and distribution of sample characteristic by study outcomes

Characteristics	School-based prevention*	Perceived second-hand smoke as harmful	Perceived tobacco as addictive	Favour smoking bans in outdoor public spaces	Favour smoking bans in enclosed public places	Would enjoy a cigarette
Full sample						
Total	59.10%	80.30%	61.0%	70.10%	74.40%	21.40%
Sex						
Male	58.10%	78.20%	57.40%	69.40%	73.20%	23.30%
Female	60.20%	82.40%	64.50%	70.90%	75.60%	19.50%
Age category						
≤12 years	54.20%	78.10%	59.90%	69.40%	74.50%	20.90%
13–15 years	60.90%	81.20%	61.10%	69.50%	74.10%	20.50%
>15 years	55.60%	78.40%	61.00%	72.50%	75.50%	24.70%
Tobacco Marketing exposure†						
No (mean)	0.84	0.89	0.88	0.9	0.84	0.87
Yes (mean)	0.98	0.93	0.95	0.93	0.95	1.12
Anti-tobacco marketing exposure‡						
No (mean)	1.02 (0.90)	1.10 (0.88)	1.20 (0.88)	1.14 (0.94)	1.08 (0.82)	1.24 (0.95)
Yes (mean)	1.42 (0.92)	1.29 (0.92)	1.29 (0.96)	1.30 (0.92)	1.30 (0.96)	1.29 (0.86)
Secondhand smoke exposure§						
No	57.90%	76.60%	56.40%	70.90%	71.10%	21.00%
Yes	60.30%	83.80%	65.20%	69.40%	77.50%	21.80%
Ever cigarette use						
No	23.2%	24.0%	21.4%	27.6%	21.8%	17.4%
Yes	21.8%	22.2%	23.4%	19.8%	21.3%	37.1%
Region						
Africa	57.10%	76.90%	52.80%	63.70%	61.80%	12.20%
Europe	64.40%	87.10%	70.10%	65.40%	79.90%	18.90%
Americas	56.40%	77.80%	61.90%	82.50%	84.20%	35.00%

*Self-reported exposure to school-based tobacco prevention programme in the past 12 months.

†Composite score (0–4) of self-reported exposure to four channels of tobacco marketing: (1) TV, movies, videos; (2) point of sale; (3) branded items and (4) offered a free tobacco product by the tobacco industry.

‡Composite score (0–3) of self-reported exposure to antitobacco marketing via: (1) media, (2) events and (3) cigarette packages.

§Self-reported exposure to secondhand smoke in the past 7 days via one or more of the following locations: (1) at home; (2) indoor public setting and (3) outdoor public setting.

or older. The mean cumulative marketing exposure for the overall sample was 0.92, and approximately 51.7% of adolescents were exposed to secondhand smoke in the past 7 days. Additionally, 6.2% of the adolescents were past 30-day cigarette smokers.

As seen in [table 2](#), approximately 59.1% of adolescents self-reported exposure to school-based tobacco prevention programming in the past 12 months.

The country-level prevalence of exposure to school-based tobacco prevention intervention ranged from 24.9% in the Comoros to 99.3% in Turkmenistan (online supplemental table 1). Additionally, 80.3% perceived

that secondhand smoke is harmful to their health, 61.0% perceived that it would be difficult to stop using tobacco, 74.4% supported a ban on tobacco use in indoor public spaces, 70.1% supported a ban on tobacco use at outdoor public spaces and 21.4% anticipated that they would enjoy cigarette smoking. A further breakdown of these variables is available in [table 2](#).

Compared with adolescents not exposed, those exposed to school-based tobacco prevention programmes were more likely to perceive secondhand smoking as harmful (74.3% vs 84.5%), perceive tobacco as addictive (55.6% vs 64.7%), support a ban on tobacco smoking in

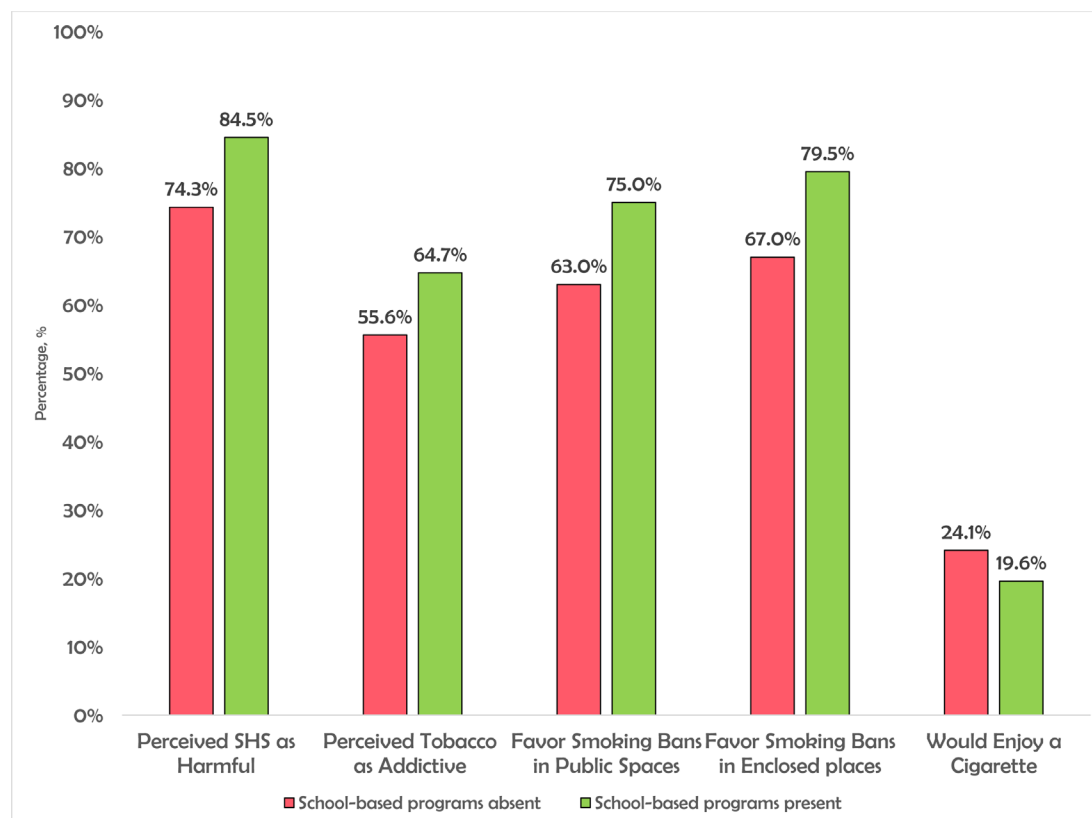


Figure 1 Tobacco-related knowledge, attitudes and behaviours, stratified by self-reported presence or absence of school-based tobacco prevention programmes among adolescents in 38 countries, 2014–2017.

outdoor public spaces (63.0% vs 75.0%) and support a ban on tobacco smoking in indoor public spaces (67.0% vs 79.5%) (figure 1). On the other hand, compared with adolescents not exposed, those exposed to school-based tobacco prevention programmes were less likely to anticipate that they would enjoy cigarette smoking (24.1% vs 19.6%).

The results from multivariate multilevel regression analyses (table 3) revealed that adolescents who self-reported exposure to school-based tobacco prevention programmes had 79% greater odds of perceiving that secondhand smoke is harmful to their health (adjusted OR (AOR): 1.79; 95% CI: 1.79 to 1.80) and 39% greater odds of perceiving that it would be difficult to stop using tobacco (AOR: 1.39; 95% CI: 1.38 to 1.39), after controlling for covariates.

Similarly, adolescents who self-reported exposure to school-based tobacco prevention programmes had greater odds of supporting a ban on tobacco use in indoor public spaces (AOR: 1.87; 95% CI: 1.87 to 1.88) and outdoor public spaces (AOR: 1.71; 95% CI: 1.71 to 1.72), controlling for covariates. And finally, adolescents who self-reported exposure to school-based tobacco prevention programmes had 23% lower odds of anticipating that they would enjoy cigarette smoking (AOR: 0.77; 95% CI: 0.76 to 0.77) after adjusting for other covariates.

DISCUSSION

This study found that approximately 2 out of 5 adolescents (40.9%) across 38 LMICs received no tobacco prevention education at school in the preceding 12 months, using GYTS data from 2014 to 2017. This finding supports assertions of a recent study that found that policies aimed at strengthening non-communicable disease prevention are poorly implemented in several LMICs.²⁵ School antismoking prevention programmes align most closely with article 12 of the WHO FCTC, given that they are geared towards exposing the population to comprehensive educational programmes on the health risks associated with tobacco use.¹⁴ Corresponding with our study, the 2019 report on the world tobacco epidemic found that this policy intervention is not widely implemented in several LMICs, particularly those in Africa.²⁶

In line with our study hypothesis, adolescents who reported exposure to school-based tobacco prevention programming reported greater tobacco harm perceptions, lower social acceptability of tobacco use and reduced positive outcome expectations related to cigarette smoking. These findings are consistent with prior literature and suggest that school-based prevention programmes may influence psychosocial factors that predict negative perceptions towards adolescent tobacco use.^{11 12 18 19} However, it should be stated that the index study is not a direct evaluation of any particular school-based tobacco prevention programmes implemented in

Table 3 Association of exposure to school-based prevention programme and psychosocial determinants of tobacco use, GYTS 2014–2017

	Perceived second-hand smoke as harmful	Perceived tobacco as addictive	Favour smoking bans in outdoor public spaces	Favour smoking bans in enclosed public places	Would enjoy a cigarette
	Adj OR	Adj OR	Adj OR	Adj OR	Adj OR
Characteristics	99% CI	99% CI	99% CI	99% CI	99% CI
School-based prevention*					
No	1.00 (referent)	1.00 (referent)	1.00 (referent)	1.00 (referent)	1.00 (referent)
Yes	1.69*** (1.69 to 1.70)	1.37*** (1.36 to 1.37)	1.59*** (1.59 to 1.60)	1.70*** (1.69 to 1.70)	0.77*** (0.76 to 0.76)
Sex					
Male	1.00 (referent)	1.00 (referent)	1.00 (referent)	1.00 (referent)	1.00 (referent)
Female	1.27*** (1.26 to 1.27)	1.29*** (1.29 to 1.30)	1.05*** (1.05 to 1.06)	1.12*** (1.11 to 1.12)	0.85*** (0.84 to 0.85)
Age					
≤12 years	1.00 (referent)	1.00 (referent)	1.00 (referent)	1.00 (referent)	1.00 (referent)
13–15 years	1.30*** (1.29 to 1.30)	1.09*** (1.09 to 1.10)	1.00 (0.99 to 1.01)	0.90*** (0.90 to 0.91)	0.83*** (0.83 to 0.84)
>15 years	1.38*** (1.38 to 1.39)	1.12*** (1.11 to 1.12)	1.13*** (1.12 to 1.13)	0.96*** (0.96 to 0.97)	0.76*** (0.75 to 0.76)
Marketing exposure†					
Mean	0.97*** (0.96 to 0.97)	1.02*** (1.02 to 1.03)	1.01*** (1.08 to 1.09)	1.00 (1.00 to 1.00)	1.27*** (1.27 to 1.28)
Anti-tobacco marketing exposure‡					
Mean	1.14*** (1.14 to 1.14)	1.03*** (1.03 to 1.03)	1.22*** (1.22 to 1.22)	1.26*** (1.26 to 1.27)	1.05*** (1.04 to 1.05)
Secondhand smoke exposure§					
No	1.00 (referent)	1.00 (referent)	1.00 (referent)	1.00 (referent)	1.00 (referent)
Yes	1.45*** (1.44 to 1.45)	1.37*** (1.37 to 1.37)	1.14*** (1.14 to 1.15)	1.47*** (1.47 to 1.48)	0.90*** (0.90 to 0.9)
Lifetime cigarette use					
No	1.00 (referent)	1.00 (referent)	1.00 (referent)	1.00 (referent)	1.00 (referent)
Yes	0.75*** (0.74 to 0.75)	0.81*** (0.81 to 0.81)	0.56*** (0.55 to 0.57)	0.60*** (0.60 to 0.60)	2.47*** (2.46 to 2.48)

Bold values reflect statistical significance (**p<0.001).

*Self-reported exposure to school-based tobacco prevention programme in the past 12 months.

†Composite score (0–4) of self-reported exposure to four channels of tobacco marketing: (1) TV, movies, videos; (2) point of sale; (3) branded items and (4) offered a free tobacco product by the tobacco industry.

‡Composite score (0–3) of self-reported exposure to antitobacco marketing via: (1) media, (2) events and (3) cigarette packages.

§Self-reported exposure to secondhand smoke in the past 7 days via one or more of the following locations: (1) at home; (2) indoor public setting and (3) outdoor public setting.

Adj OR, adjusted OR; GYTS, Global Youth Tobacco Survey.

these LMICs (ie, their curriculum or implementation procedures).

Despite the positive association between school prevention programmes and the psychosocial predictors of tobacco use examined in this study, findings revealed that school-based prevention programmes are not widespread in LMICs. This highlights an important opportunity to

implement effective tobacco control intervention within an at-risk population. Besides supporting the implementation of article 12 of the WHO FCTC, school-based prevention programmes can help cultivate a generation of well-informed tobacco control advocates. Most smokers initiate tobacco use prior to the age of 18 years,^{8 13} hence, effective implementation of smoking prevention

programmes focused on adolescents can substantially cut the number of newly recruited smokers in each country. Therefore, the absence of these programmes in schools across LMICs further represents a missed opportunity to curb smoking initiation among adolescents.

This study builds on the large pool of literature that indicates a need for robust, comprehensive and equitably available school-based programmes to address adolescent tobacco use. As such, regulatory action should be considered that assist in developing sustainable funding for school-based prevention programmes. Similarly, LMICs may consider setting user fees for tobacco manufacturing and licensing for tobacco retailers, as these strategies have been used in other regions of the world to develop and sustain tobacco use interventions.^{13 27–29} While several school-based prevention programmes have been implemented in schools in other regions of the world, adolescents in LMICs have not received equitable engagements to address the high prevalence of tobacco use among adolescents.^{11–13 30–32}

For national and local public health agencies, this study highlights the importance of developing, implementing and evaluating school-based tobacco prevention programmes as a means to reducing adolescent tobacco use in LMICs. More so, it must be noted that school prevention programmes may be more successful in curbing adolescent tobacco use when implemented together with other tobacco control policies as stipulated by the WHO FCTC and measured by the MPOWER framework. For example, complementing school prevention programmes with comprehensive enforcement of complete national bans on tobacco advertising promotion and sponsorship, as well as bans on sales of tobacco products to minors, could achieve desired results than implementing school-based tobacco intervention programmes in isolation. While funding school smoking prevention programmes may present a major bottleneck to countries in LMIC, strategies entailing empowering judicial actors in LMIC to hold the tobacco industry financially responsible for implementing antismoking programmes have been proposed and could be expanded to school prevention programmes.³³

This study has limitations. First, all data are cross-sectional; thus, we cannot speak to any causal or temporal relationships between exposure to school-based tobacco prevention programming and psychosocial determinants of tobacco use among adolescents in LMICs. Second, while this study did examine psychosocial constructs (eg, harm perceptions; outcome expectations) associated with tobacco use during adolescence, the measures available for this study were limited in psychometric robustness. Thus, more direct research is needed. Given these first two limitations, this study should not be interpreted as a direct examination of the effectiveness and reach of school-based tobacco prevention programmes in LMICs but rather as an exploratory assessment in the assessed regions. Also, the study is prone to recall bias, given that participants had to recollect their exposure to tobacco

prevention interventions in the past; recall bias is also present in the assessment of self-reported marketing exposure. In addition, the sample countries are not representative of all countries in the respective regions; results are therefore not generalisable to the regions. However, given the high response rates of GYTS in each country studied, prevalence estimates are nationally representative. There is also potential for misreporting, as the target age group for GYTS are students in grade levels corresponding to ages 13–15 years. These early teens may not be aware of the full spectrum of interventions that constitute tobacco prevention activities and may under-report their presence. For example, the survey focused on only curriculum-based interventions, but did not capture others, such as peer-led or non-didactic/social interventions. Lastly, our study did not take into account the variability in age of exposure to tobacco educational programmes occasioned by the differences in the structure and form of educational curricula as well as the age of school enrolment among the various countries and regions.

CONCLUSION

This study found that exposure to school-based tobacco prevention programmes was associated with increased adolescents' perceived harm from secondhand smoking and addiction to nicotine and increased support for restrictions on smoking in indoor and outdoor public spaces. Exposure to school-based tobacco prevention programmes was associated with decreased expectations of enjoying smoking among adolescents. National governments should incorporate school-based education programmes into ongoing tobacco control strategies to curb cigarette smoking among adolescents.

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